

Bi-Level Control of Weaving Sections in Mixed Traffic Environments With Connected and Automated Vehicles

Longhao Yan¹, Member, IEEE, Jinhao Liang¹, and Kaidi Yang¹, Member, IEEE

Abstract—Connected and automated vehicles (CAVs) can be beneficial for improving the operation of highway bottlenecks such as weaving sections. This paper proposes a bi-level control approach based on an upper-level deep reinforcement learning controller and a lower-level model predictive controller to coordinate the lane-changings of a mixed fleet of CAVs and human-driven vehicles (HVs) in weaving sections. The upper level represents a roadside controller that collects vehicular information from the entire weaving section and determines the control weights used in the lower-level controller. The lower level is implemented within each CAV, which takes the control weights from the upper-level controller and generates the acceleration and steering angle for individual CAVs based on the local situation. The lower-level controller further incorporates an HV trajectory predictor, which is capable of handling the dynamic topology of vehicles in weaving scenarios with intensive mandatory lane changes. The case study inspired by a real weaving section in Basel, Switzerland, shows that our method consistently outperforms state-of-the-art benchmarks.

Index Terms—Weaving sections, connected and automated vehicles, mixed traffic, lane-changing, model predictive control, deep reinforcement learning.

I. INTRODUCTION

WEAVING sections are highway segments where merges are located close to diverges. Due to intense lane-changing maneuvers between the auxiliary and main lanes, weaving sections are recognized as key highway bottlenecks, characterized by a phenomenon called capacity drop. Such a capacity drop phenomenon has been widely reported in the literature, whereby the discharge flow of the weaving section can be reduced by 3%-20% after the onset of congestion [1], [2], further deteriorating highway traffic operations. Although researchers have identified potential causes of capacity drop (e.g., waste in space, undesirable distribution of lane-changing positions, etc.) and proposed various methods to mitigate capacity drop [3], [4], these works focus on scenarios with only human-driven vehicles (HVs), which might be

ill-informed due to limited perception capability [5] and act independently with minimal coordination [6].

The development of connected and automated vehicles (CAVs) holds promise in improving traffic operations in weaving sections [7], [8], [9], [10]. Enabled by vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) wireless communications, CAVs can provide holistic and predictive information about the weaving section. Moreover, unlike HVs that are controlled by independent drivers, CAVs can accurately follow instructions provided by a central controller and thus can be coordinated to improve system-wide efficiency in weaving sections.

To exploit the benefits of CAVs in highway systems, researchers have proposed various methods to control lane-changing maneuvers [11], [12], [13], [14]. Most of these works assume no coordination or only local coordination that only coordinates a vehicle with its neighbors [15], [16], [17]. For example, Ref. [15] developed DRL algorithms to make car-following and lane-changing decisions for a single vehicle while no coordination is applied among vehicles. Ref. [16] proposed a multi-agent DRL-based lane-changing control approach in which vehicles are controlled in a decentralized manner using only information collected by themselves. Ref. [17] proposed a harmonious deep reinforcement learning (DRL)-based algorithm to facilitate lane changing for CAVs with limited perception range, whereby all vehicles act simultaneously to make independent decisions to balance system and individual efficiency. Although these local coordination approaches can be computationally efficient in improving highway operations by involving only a few vehicles, they focus only on a small portion of highway segments, such as on-ramps and off-ramps [18], [19], [20], [21], and therefore ignore the global impact of the resulting local decisions on the entire weaving section, making it difficult to achieve system-wide efficiency. In fact, globally coordinated control is argued to be essential for improving traffic efficiency in weaving sections, particularly on the mainline, where mandatory lane changes are prevalent [22], [23].

To the best of our knowledge, only a small portion of studies have developed *global coordination* methods for jointly controlling vehicles across weaving sections [8], [24], [25], [26], [27]. Ref. [23] proposed a dynamic programming algorithm to coordinate the trajectory planning of all CAVs in scenarios with intensive mandatory lane-changing maneuvers.

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Ref. [28] developed a hybrid Model Predictive Control (MPC) method for systematic cooperative lane-changing control to improve traffic smoothness and driving comfort. A distributed cooperative lane-changing method was developed by [22], which utilized the predictive information from other vehicles to solve a mixed integer optimization problem. Ref. [27] proposed a coalitional game theory approach for cooperative lane-changing decisions to avoid the win-lose situations that occur in individual decision-making. However, these aforementioned works suffer from two major limitations. First, they often require solving complex optimization problems that scale poorly with increasing vehicle numbers, especially for large-scale coordination across the entire weaving section. Second, most of these works assume all vehicles to be CAVs without considering the transition period in which CAVs are gradually developed and deployed. Although Ref. [3] considered such a transition period, this work leveraged a macroscopic traffic modeling approach to derive the optimal distribution of lane-changing positions without coordinating individual CAVs.

Statement of Contribution: To address the issues with existing local and global coordination methods, we propose a computationally efficient bi-level controller that combines DRL and MPC to globally coordinate lane-changing decisions of all CAVs in mixed-traffic weaving scenarios. Our contribution is two-fold. First, to address the research gap of lacking computationally efficient global CAV coordination algorithms for mixed-traffic weaving sections, we propose a bi-level controller with an upper-level DRL and a lower-level MPC. The upper-level DRL-based controller represents a roadside coordinator that collects information from each vehicle to decide the weights of multiple control objectives, which are used to inform the lower-level controller. The lower-level MPC-based controller is implemented in each vehicle, which generates driving strategies for individual CAVs in local situations by solving a simple quadratic optimization problem with an objective function parametrized by upper-level decisions. Our bi-level approach can enhance system-wide efficiency while allowing individual vehicles to react to changing local situations considering safety and adaptability in mixed-traffic scenarios. Second, unlike a large number of MPC-based controllers that assume constant speed or pre-known trajectory for surrounding HVs [29], [30], at the lower level, we design an HV trajectory prediction algorithm that can predict the trajectories of multiple HVs around a CAV. Specifically, we leverage EvolveGCN [31] to consider the evolution of the dynamic topology of surrounding HVs around CAV. The predicted trajectories provide necessary input to the lower-level MPC-based controller.

The remainder of this paper is organized as follows. Section II describes the problem formulation and overall framework of the proposed bi-level controller. Section III describes the upper-level DRL-based controller, and Section IV elaborates the formulation of MPC-based controller and the details of HV trajectory predictor. Section V performs the simulation of a typical weaving section. Section VI concludes the paper and presents future work.

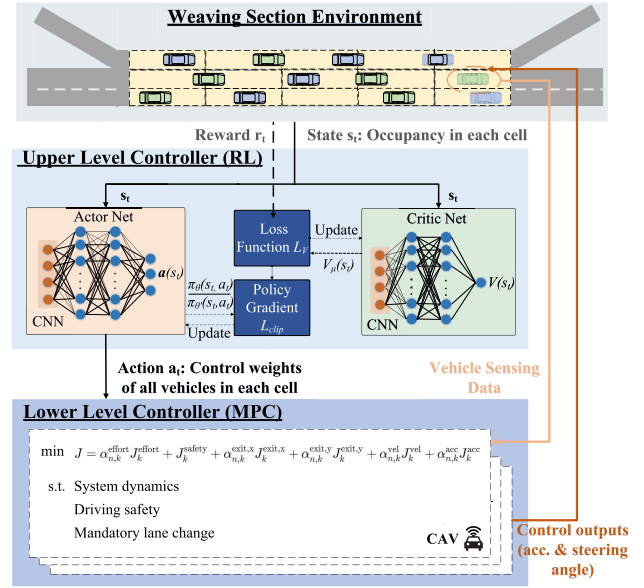


Fig. 1. Framework of the global coordinated bi-level controller in weaving sections. The studied area is marked yellow in the weaving section environment. The upper-level controller is based on Reinforcement Learning (RL). The lower-level controller is based on Model Predictive Control (MPC).

II. PROBLEM FORMULATION AND OVERALL FRAMEWORK

We consider a typical weaving section of L lanes, with an on-ramp closely located to an off-ramp, where the trajectories of vehicles form a weaving pattern. Note that weaving trajectories make weaving sections differ from on-ramp scenarios where vehicles only change from on-ramp to highway mainline and from off-ramps where lane-change density is much lower due to the sufficient space for changing lanes.

We further consider scenarios with mixed traffic of CAVs and HVs, whereby CAVs are controlled by their onboard controllers, and HVs are controlled by drivers. Both CAVs and HVs are assumed to be connected and can communicate with a roadside unit installed near the weaving section through V2I communication. We make this assumption because we envision that vehicle connectivity will be prevalent when automated vehicles are deployed. In fact, the Land Transport Authority in Singapore is mandating Singapore-registered vehicles to install onboard units, making almost all vehicles connected in the near future. Mathematically, let $\mathcal{C}$ represent the set of CAVs, and $\mathcal{H}$ represent the set of HVs. The state of each vehicle $i \in \mathcal{C} \cup \mathcal{H}$ (either CAV or HV) is represented by a tuple $s_i = (e_i, \mathbf{p}_i, v_i, \psi_i)$, with e_i indicating its destination lane, $\mathbf{p}_i = (p_{x,i}, p_{y,i})$ the position vector containing the longitudinal coordinate $p_{x,i}$ and the lateral coordinate $p_{y,i}$, v_i the longitudinal speed, and ψ_i the heading angle. The considered time horizon is discretized into a set $\mathcal{K}$ of time intervals (indexed by k) with size Δt .

The goal of this work is to develop a global coordination algorithm for CAVs to improve the operation of the weaving sections in mixed traffic. To this end, we propose a bi-level control approach, as demonstrated in Fig. 1, whereby the roadside unit implements an RL-based upper-level controller

(see Section III), and each CAV implements an MPC-based lower-level controller (see Section IV). The upper-level controller collects information from all vehicles on the road and then determines the weights of various control objectives (hereafter referred to as control weights) in lower-level controllers to optimize system performance. The lower-level controller embedded in each CAV first predicts the trajectories of the surrounding HVs, if any, and then determines the ego CAV's acceleration rate and steering angle to optimize various control objectives, such as reaching the target, traffic efficiency, and safety, using the control weights calculated at the upper level.

We make the following three remarks regarding the bi-level framework. First, the upper-level controller coordinates the lane-changing positions of vehicles by tuning the control weights. These control weights have a clear physical meaning, which is closely related to the positions of CAVs. As the CAV approaches the exit point, the lane changing becomes more urgent, so the corresponding weight for reaching the target should be higher. Second, the proposed bi-level controller can improve mixed-traffic weaving section operations by achieving the global coordination of CAVs in a computationally efficient manner while providing local safety guarantees to CAVs. In comparison, it can be challenging for pure RL algorithms to handle a large and dynamically changing number of agents and ensure the safety of individual vehicles. It can also be challenging for MPC-based approaches to facilitate the global coordination of CAVs. Third, for presentation simplicity, we assume both levels make decisions at the same frequency (i.e., with the same time interval Δt). This assumption can be readily relaxed such that the upper-level controller makes decisions at a lower frequency than the lower-level controller (see the sensitivity analysis in Section V-E).

We further make four assumptions regarding the driving behavior of vehicles in weaving sections. First, we make a realistic assumption that the HV driving behavior is heterogeneous and unknown to CAVs. Second, we assume that HVs do not adapt their driving strategies to the real-time control of CAVs because such adaptation can only happen slowly after HVs obtain sufficient observations about CAV behavior. Third, we assume that vehicles would position themselves in the lane closest to their desired lane before entering the weaving section, as competitive lane changes happen in weaving sections, and drivers are unwilling to exit from an undesired lane. Fourth, we assume CAVs only perform mandatory lane changes within the weaving section, i.e., from the mainline to the off-ramp (diverging) and from the on-ramp to the mainline (merging), which are essential for vehicles to continue along their intended routes. CAVs are not permitted to perform discretionary lane changes within the intersection. This assumption is made because mandatory lane changes are widely acknowledged as a dominating driving behavior and primary contributor to the capacity drop phenomenon [3], [23], [32]. Moreover, as this assumption provides restrictions on CAV movements, our results provide a conservative (i.e., lower-bound) estimate of achievable performance. The relaxation of this assumption will be performed in future work.

III. RL-BASED UPPER-LEVEL CONTROLLER FOR GLOBAL COORDINATION

We formulate the decision-making process of the upper-level controller (i.e., the roadside controller) as a Markov Decision Process (MDP) $\mathcal{T} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, R, \gamma)$, where $\mathcal{S}$ represents the state space, $\mathcal{A}$ the action space, $\mathcal{P}$ system dynamics, R reward function, and γ a scalar discount factor.

State Space $\mathcal{S}$: Let us divide the lanes of the weaving section into a set of cells, collectively denoted by $\mathcal{N}$ and indexed by n . We define the states at time step $k \in \mathcal{K}$ as $s_k = \{g_{n,k}\}_{n \in \mathcal{N}}$, where $g_{n,k}$ represents the state of cell $n \in \mathcal{N}$ of the past D time steps, represented as $g_{n,k} = \{l_{n,k-h}, b_{n,k-h}, \bar{v}_{n,k-h}\}_{h=0}^{D-1}$, which includes the lane change occupancy $l_{n,k-h}$ (i.e., the ratio of vehicles not on the destination lane), the number of vehicles located in the cell $b_{n,k-h}$, and the average speed $\bar{v}_{n,k-h}$ of all vehicles within cell n at time step $k-h$.

Action Space: We define the upper-level action as $\alpha = \{\alpha_{n,k}\}_{n \in \mathcal{N}}$, where $\alpha_{n,k} = \{\alpha_{n,k}^{\text{effort}}, \alpha_{n,k}^{\text{exit,x}}, \alpha_{n,k}^{\text{exit,y}}, \alpha_{n,k}^{\text{vel}}, \alpha_{n,k}^{\text{acc}}\}$ represents the control weights in the MPC-based lower-level controller (see Section IV for details) for all CAVs in cell n , which penalizes the control effort ($\alpha_{n,k}^{\text{effort}}$), the distance to the target point ($\alpha_{n,k}^{\text{exit,x}}$ and $\alpha_{n,k}^{\text{exit,y}}$), the deviation from the desired speed ($\alpha_{n,k}^{\text{vel}}$), and the impact on surrounding vehicles ($\alpha_{n,k}^{\text{acc}}$). We make three remarks about using these control weights as upper-level actions. First, these weights are used by every CAV in cell n to trade off these control objectives to adapt the lower-level controller to various traffic conditions. Second, these control weights can help achieve global coordination in that they can encode the global information about the weaving section through the policy network trained in the RL-based upper-level controller.

System Dynamics: We describe the system dynamics $\mathcal{P}$ as the traffic dynamics of the weaving section, including the arrival and departure of vehicles, the car-following and lane-changing decisions of the HVs, and the kinematic models of both the HVs and CAVs. Note that the system dynamics are unknown from the perspective of the upper-level RL controller.

Reward Function: We define the reward r_k^{total} at time step k as the weighted sum of four components, including (i) a speed component r_k^{spd} , (ii) a traffic flow component r_k^{flow} , (iii) a missing penalty component r_k^{miss} , and (iv) a safety component r_k^{safety} .

$$r_k^{\text{total}} = \beta^{\text{spd}} r_k^{\text{spd}} + \beta^{\text{flow}} r_k^{\text{flow}} + \beta^{\text{miss}} r_k^{\text{miss}} + \beta^{\text{safety}} r_k^{\text{safety}} \quad (1)$$

where β^{spd} , β^{flow} , β^{miss} and β^{safety} are the corresponding weights defined by the traffic operator. The specific forms of these three reward components are given below:

$$\begin{aligned} r_k^{\text{spd}} &= \frac{1}{N_k} \sum_{i=1}^{N_k} \frac{v_{i,k}}{v_f}, \quad r_k^{\text{flow}} = \frac{q_{D,k}}{q_{\max}}, \\ r_k^{\text{miss}} &= b_k^{\text{miss}}, \quad r_k^{\text{safety}} = \frac{1}{N_k} \sum_{i=1}^{N_k} \frac{\zeta_{i,k}}{d_{\text{safety}}} \end{aligned} \quad (2)$$

where the speed component r_k^{spd} represents the normalized space-mean speed on the weaving section, with N_k , $v_{i,k}$, and v_f indicating the number of vehicles on the weaving section

at time step k , the speed of vehicle i at time step k , and the free-flow speed, respectively. Notice that the calculation of r_k^{spd} considers both CAVs and HVs to ensure that the decisions made by RL benefit the entire traffic flow regardless of vehicle type. The traffic flow component r_k^{flow} represents the exit flow $q_{D,k} = \{q_{k-h}\}_{h=0}^{D-1}$ of the weaving section in the past D time steps, normalized by the theoretical capacity $q_{\max}$. Here, the theoretical capacity refers to the capacity without considering lane changing, as frequent lane changes in weaving sections can lead to capacity drop. The missing penalty component r_k^{miss} is defined to be the number of vehicles that leave the weaving section without successfully changing to their destination lane at time step k , represented by b_k^{miss} . The safety component r_k^{safety} represents the violation of the safety distance $\zeta_{i,k}$, normalized by the predefined safety distance d^{safety} .

Deep Reinforcement Learning Algorithm: The formulated MDP is solved using a DRL algorithm, Proximal Policy Optimization (PPO), which is selected due to its stability and wide application in traffic and vehicle control [11], [33].

PPO leverages an actor-critic structure, whereby the actor network takes the grid-like global information of the weaving section as input and provides an output of the control weights for the lower-level MPC controller, and the critic network takes the same input but outputs the estimated state value of MDP. Both the actor and the critic network use Convolutional Neural Networks (CNN) as input layers to handle grid-like states [34].

The actor network determines a policy $\pi_\theta(\alpha | s)$ parameterized by θ to minimize a clipped loss represented in Eq.(3), where a clip function with threshold parameter κ is used to avoid excessive updates that could result in performance oscillations:

$$L_{\text{CLIP}}(\theta) = \mathbb{E} \left[\min \left(\hat{A}_k \frac{\pi_\theta(\alpha | s)}{\pi_{\theta_{\text{old}}}(\alpha | s)}, \hat{A}_k \text{clip} \left(\frac{\pi_\theta(\alpha | s)}{\pi_{\theta_{\text{old}}}(\alpha | s)}, 1 - \kappa, 1 + \kappa \right) \right) \right] \quad (3)$$

where $\pi_{\theta_{\text{old}}}$ is an old policy updated regularly to confine the variance of the current policy, and $\hat{A}_k$ is an advantage function to accelerate convergence.

The critic network estimates the state value $V_\mu(s_k)$ of the MDP. The loss function of critic net with parameter μ is defined as:

$$L^V(\mu) = \mathbb{E} \left[\left(\hat{V}_\mu^{\text{target}}(s_k) - V_\mu(s_k) \right)^2 \right] \quad (4)$$

where $\hat{V}_\mu^{\text{target}}(s_k) = r_{k+1} + \gamma V_\mu(s_{k+1})$ is the state value that $V_\mu(s_k)$ aims to approach.

The actions determined by the DRL-based upper-level controller will be leveraged to inform the MPC-based lower-level controller, as presented next.

IV. MPC-BASED LOWER-LEVEL CONTROLLER FOR INDIVIDUAL VEHICLE

In this section, we first introduce our proposed lower-level MPC controller that combines behavioral decision-making, path planning, and motion control of individual CAVs in Section IV-A. Then, we present our HV trajectory prediction algorithm to generate input for the MPC in Section IV-B.

A. MPC Formulation

In this subsection, we present the lower-level controller for CAV $i \in \mathcal{C}$ with its center of gravity (COG) located within cell n_i on lane $o_{i,k}$ with destination lane e_i . For brevity, we omit the index i in this subsection where its omission does not introduce ambiguity. At each time step k , this CAV i , referred to as the subject vehicle (SV), determines its optimal lane-changing decision, acceleration rate, and steering angle using two types of input: (i) the detected states of its surrounding vehicles within a sensing range and (ii) the control weights associated with the cell in which it is located, which is received from the upper-level controller. These control decisions are made through an MPC framework, where optimization is performed to minimize total costs within a planning horizon of the next P time steps, but only the one corresponding to the current time step is executed. The embedded optimization problem of the MPC framework is a convex optimization problem that minimizes a combination of control objectives:

$$J = \alpha_{n,k}^{\text{effort}} J_k^{\text{effort}} + J_k^{\text{safety}} + \alpha_{n,k}^{\text{exit,x}} J_k^{\text{exit,x}} + \alpha_{n,k}^{\text{exit,y}} J_k^{\text{exit,y}} + \alpha_{n,k}^{\text{vel}} J_k^{\text{vel}} + \alpha_{n,k}^{\text{acc}} J_k^{\text{acc}} \quad (5)$$

where J_k^{effort} , J_k^{safety} , $J_k^{\text{exit,x}}$, $J_k^{\text{exit,y}}$, J_k^{vel} , and J_k^{acc} , respectively, represent the control effort, safety penalty, longitudinal distance to the target point, lateral distance to the target point, deviation from the desired speed, and the impact on the surrounding vehicles at time step k , weighted by the control weights (relative to the safety penalty) obtained from the upper-level controller $\alpha_{n,k} = \{\alpha_{n,k}^{\text{effort}}, 1, \alpha_{n,k}^{\text{exit,x}}, \alpha_{n,k}^{\text{exit,y}}, \alpha_{n,k}^{\text{vel}}, \alpha_{n,k}^{\text{acc}}\}$.

Notation: We use $z(k)$ to represent the value of variable z at time step k observed in the real system, $z(k+t|k)$ to represent the predicted value of variable z at time step $k+t$ with the prediction being made at time step k , and $\mathbf{z}_k = \{z(k+t|k)\}_t$ to represent the vector of variables formed by $z(k+t|k)$.

Next, we present the details of these control objectives and the corresponding constraints.

1) **System Dynamic:** We first describe the vehicle dynamics and obtain the control objective of the control effort J_k^{effort} . For vehicle dynamics, we employ the widely adopted bicycle model [35] to describe the dynamics of longitudinal position p_x , lateral position p_y , heading angle ψ , and longitudinal speed v of SV.

$$\dot{p}_x = v \quad (6a)$$

$$\dot{p}_y = v\psi + \frac{l_r}{l_f + l_r} v\delta_f \quad (6b)$$

$$\dot{\psi} = \frac{l}{l_f + l_r} v\delta_f \quad (6c)$$

$$\dot{v} = a_x \quad (6d)$$

where a_x and δ_f are the longitudinal acceleration and steering angle, respectively, serving as the decision variables. l , l_f , and l_r represent the length of the axle, the distance of the front axle from the COG, and the distance of the rear axle from the COG, respectively.

To facilitate the design of MPC, we linearize and discretize the bicycle model Eq.(6a)- Eq.(6d) into a discrete-time linear

time-invariant system as follows.

$$\mathbf{x}(k+t+1|k) = \mathbf{A}\mathbf{x}(k+t|k) + \mathbf{B}\mathbf{u}(k+t|k) \quad (7)$$

where the state variable $\mathbf{x}(k+t|k) = [p_x(k+t|k), p_y(k+t|k), \psi(k+t|k), v(k+t|k)]$, control variable $\mathbf{u}(k+t|k) = [a_x(k+t|k), \delta_f(k+t|k)]$. Matrices $\mathbf{A}$ and $\mathbf{B}$ describe the linearized system dynamics.

Furthermore, we link SV's lateral positions with the lane-changing decisions. Since intensive lane changes can reduce traffic efficiency, we only allow SV to perform mandatory lane changes towards its destination lane e . Moreover, since consecutive lane changes within a short time period are typically not allowed, we assume that during the planning horizon, SV can only change to the adjacent lane in the direction of its destination lane e . In particular, we use a binary variable $\varsigma_k \in \{0, 1\}$ to indicate whether lane changes will happen during the planning horizon starting at time step k , with $\varsigma_k = 1$ indicating lane changing. Accordingly, the above requirements can be mathematically represented as follows:

$$\varsigma_k \leq |o_k - e| \quad (8)$$

$$\varsigma_k \geq \varsigma_{k-1}(1 - |o_k - o_{k-1}|) \quad (9)$$

$$p_y(k+P|k) = \begin{cases} p_{y,o_k}, & \text{if } o_k = e \\ \varsigma_k p_{y,o_k+1} + (1 - \varsigma_k) p_{y,o_k}, & \text{if } o_k < e \\ \varsigma_k p_{y,o_k-1} + (1 - \varsigma_k) p_{y,o_k}, & \text{if } o_k > e \end{cases} \quad (10)$$

$$\begin{cases} p_{y,o_k}^- \leq p_y(k+t|k) \leq p_{y,o_k}^+, & \text{if } o_k = e \\ p_{y,o_k}^- \leq p_y(k+t|k) \leq \varsigma_k p_{y,o_k+1}^+ + (1 - \varsigma_k) p_{y,o_k}^+, & \text{if } o_k < e \\ \varsigma_k p_{y,o_k-1}^- + (1 - \varsigma_k) p_{y,o_k}^- \leq p_y(k+t|k) \leq p_{y,o_k}^+, & \text{if } o_k > e \end{cases} \quad (11)$$

where Eq.(8) represents that lane changing decision can only be made when the current lane o_k does not coincide with the destination lane e (i.e., $o_k \neq e$). Eq.(9) represents that if lane changing is made at the previous time step $k-1$ and the lane change is not completed, the lane changing process will continue at time step k . Eq.(10) calculates the lateral position at the end of the planned horizon, $p_y(k+P|k)$, where p_{y,o_k} represents the lateral position of the center line of lane o_k on which SV is currently located. If SV decides not to change lane ($\varsigma_k = 0$), its lateral position at the end of the prediction horizon, $p_y(k+P|k)$, would be the center of its current lane. If SV decides to change lane ($\varsigma_k = 1$), its lateral position at the end of the prediction horizon, $p_y(k+P|k)$, would be the center of its adjacent lane (p_{y,o_k+1} or p_{y,o_k-1}). Eq.(11) represents the range of the lateral positions of SV during the planning horizon (with $t = 1, \dots, P$), where $p_{y,\xi}^-$ and $p_{y,\xi}^+$ represent the lateral position of lane ξ 's borders. Specifically, the SV's lateral position during the planning horizon should be within the borders of its current lane if the lane-keeping decision is made ($\varsigma_k = 0$) and between its current lane and its adjacent lane in the direction of its destination lane if the lane-changing decision is made ($\varsigma_k = 1$).

Next, we define the physical limitation constraints for other state and control variables of SV within the planned horizon. First, SV's speed should be upper-bounded by the speed limit

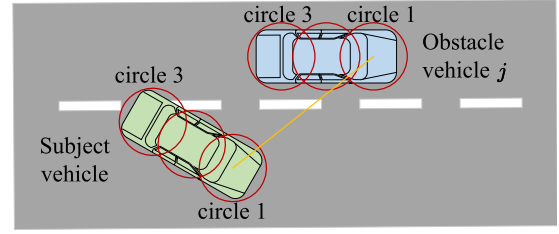


Fig. 2. Circle decomposition for measuring inter-vehicle distance.

on the weaving section. Second, SV's acceleration rate and steering angle are constrained by the operational limits defined by the vehicle's mechanics. Overall, these physical limitation constraints are summarized as follows:

$$\mathbf{x}_{\min} \leq \mathbf{x}(k+t|k) \leq \mathbf{x}_{\max} \quad (12)$$

$$\mathbf{u}_{\min} \leq \mathbf{u}(k+t|k) \leq \mathbf{u}_{\max} \quad (13)$$

where $\mathbf{x}_{\min}$ and $\mathbf{x}_{\max}$ represent the lower and upper bounds for state variables, respectively, and $\mathbf{u}_{\min}$ and $\mathbf{u}_{\max}$ represent the lower and upper bounds for the control variables, respectively.

Then, the control effort J_k^{effort} describes the energy consumption associated with the control variables, represented as follows:

$$J_k^{\text{effort}} = \sum_{t=1}^P \|\mathbf{u}(k+t|k)\|_2^2 \quad (14)$$

2) *Driving Safety*: We now establish the driving safety constraints and derive the safety penalty J_k^{safety} . We consider driving safety as keeping the distance between a CAV and its surrounding vehicles beyond a predefined safety threshold d^{safety} . As the MPC operates in a moving horizon manner, it is sufficient to consider the surrounding vehicles in direct conflict with the SV, denoted by set $\mathcal{M}$. In highway driving scenarios, the set $\mathcal{M}$ can include the leading and following vehicles in the current lane, vehicles in adjacent lanes to cut in before the SV, and leading and following vehicles in the adjacent lane in the direction of the SV's destination lane (if the SV is performing lane-changing). Moreover, we only consider surrounding vehicles that can be continuously observed by the SV for more than three seconds. This is a practical requirement because vehicles outside of three seconds are impossible to collide with the SV.

Due to the complex shape of vehicles, directly calculating the distance between the SV and a surrounding vehicle can be challenging. To address this, we follow [36] and approximate each vehicle by three circles centered at its front axle, center of gravity, and rear axle, respectively, as illustrated in Fig. 2. The inter-vehicle distance is then calculated as the minimum of the distances between these centers. The safety constraints and the safety penalty component in the objective function can be written as follows:

$$\begin{aligned} \|\mathbf{p}_l^{\text{SV}}(k+t|k) - \mathbf{p}_{j,w}^{\text{safety}}(k+t|k)\|_2 &\geq d^{\text{safety}} - \iota_{jw}(k+t|k), \\ \forall j \in \mathcal{M}, l \in \{1, 2, 3\}, w \in \{1, 2, 3\}, t \in \{1, \dots, P\} \end{aligned} \quad (15)$$

$$J_k^{\text{safety}} = \sum_{t=1}^P \chi^t \|\zeta_t\|_2^2 \quad (16)$$

where Eq.(15) represents that the inter-vehicle distance (left-hand side) is lower-bounded by a threshold, with p_l^{SV} indicating the center position of SV's l -th circle, $p_{j,w}^{\text{safety}}$ indicating the center position of w -th circle of surrounding vehicle $j \in \mathcal{M}$. The relaxation term u_{ljw} indicates the violation between SV's l -th circle and w -th circle of surrounding vehicle j . This term is penalized in the objective function Eq.(5) via the safety penalty component Eq.(16), which converts hard safety constraints to soft constraints to ensure the feasibility of the embedded optimization problem of MPC. In Eq.(16), $\zeta_t = \sum_{l=1}^3 \sum_{w=1}^3 \sum_{j \in \mathcal{M}} u_{ljw}(k+t|k)$ represents the cumulative violations of safe distance between SV and its surrounding vehicles in the set $\mathcal{M}$, and the decay factor χ is designed to highlight the risks of immediate collisions.

If the surrounding vehicle j is an HV, the values of $p_{j,w}^{\text{safety}}(k+t|k)$ can be calculated from the predicted position of the surrounding vehicle j , which can be obtained using the trajectory prediction algorithm presented in Section IV-B. If surrounding vehicle j is a CAV, the value of $p_{j,w}^{\text{safety}}$ can be inferred from the control variables of vehicle j at time step $k-1$ obtained through V2V communication.

Note that the established formulation Eq. (15) is nonlinear, which makes the entire optimization problem hard to solve. To improve computation efficiency, we linearize Eq. (15) using its first-order Taylor expansion as suggested in [37].

$$A_k^{\text{safety}} [(U_k)^T, (X_k)^T, (Z_k)^T]^T \leq b_k^{\text{safety}} \quad (17)$$

$$-Z_k \leq 0 \quad (18)$$

where $U_k = \{u(k+t|k)\}_{t=0}^{P-1}$, $X_k = \{x(k+t|k)\}_{t=1}^P$, and $Z_k = \{u_{ljw}(k+t|k)\}_{l \in \{1, \dots, P\}, w, l \in \{1, 2, 3\}, j \in \mathcal{M}}$ are the vectors of the control, state, and slack variables, respectively. Matrix A^{safety} and vector b^{safety} define the linearized constraint.

3) *Exit Attraction*: Next, we calculate $J_k^{\text{exit},x}$ and $J_k^{\text{exit},y}$, which represent the exit attraction defined as the vehicle's desire to approach its destination exit point. Such criteria are especially important for weaving sections where most lane changes are mandatory. As the SV approaches the end of the weaving section, its urgency in performing lane change increases. Consequently, we can define the cost terms $J_k^{\text{exit},x}$ and $J_k^{\text{exit},y}$ as follows:

$$J_k^{\text{exit},x} = (p_x(k+P|k) - \hat{p}_{x,k})^2 \quad (19a)$$

$$J_k^{\text{exit},y} = (p_y(k+P|k) - p_{y,d})^2 \quad (19b)$$

where $p_{y,d}$ represents the coordinate of the lateral center of the destination lane, and $\hat{p}_{x,k}$ represents SV's estimated longitudinal position at the end of the prediction horizon. The value of $\hat{p}_{x,k}$ is estimated by assuming a free-flow speed v_f throughout the prediction period.

4) *Target Speed Attraction*: To encourage the SV to maintain a high speed, an attractive force toward the target speed is defined as follows:

$$J_k^{\text{vel}} = (v_k - v_f)^2 \quad (20)$$

where the target speed is set to the free-flow speed v_f along the weaving section.

5) *Impact on SV and Other Vehicles*: We finally account for the impact of SV's lane change on surrounding vehicles to obtain J_k^{acc} using an incentive term inspired by the Minimize Overall Braking Induced by Lane Change (MOBIL) model [38], which calculates the impact of SV's lane changing decision on itself and surrounding vehicles:

$$B_k^{\text{acc}} = \underbrace{a_{i,k} - \hat{a}_{i,k}}_{\text{SV}} + \rho \underbrace{(a_{f,k} - \hat{a}_{f,k})}_{\text{new follower}} + \underbrace{a_{f',k} - \hat{a}_{f',k}}_{\text{old follower}} \quad (21)$$

where a and $\hat{a}$ represent longitudinal acceleration rates before and after lane change, respectively, and the subscripts i , f , and f' indicate SV, the new follower after lane change, and the old follower before lane change, respectively. ρ is the factor called politeness that represents the level of consideration of the surrounding vehicles.

To estimate the acceleration after lane change, we leverage the Intelligent Driver Model (IDM) [39], which models the longitudinal behavior of a vehicle as:

$$\dot{v} = a_{\max} \left[1 - \left(\frac{v}{v_f} \right)^4 - \left(\frac{d^*}{d} \right)^2 \right] \quad (22)$$

$$d^* = d_{\text{jam}} + \tau v + \frac{v \Delta v}{2 \sqrt{a_{\max} a_{\min}}} \quad (23)$$

where v is the vehicle speed, d is the spacing, and Δv is the speed difference between the vehicle and its leading vehicle. Parameter τ is the desired time headway, d_{jam} is the jam spacing, and $a_{\max}$ and $a_{\min}$ are the maximum and minimum acceleration rates, respectively. Then, the acceleration rates $\hat{a}_{i,k}$, $\hat{a}_{f,k}$, and $\hat{a}_{f',k}$ in Eq. (21) can be computed by Eq. (22) and (23) at time step k .

Then, we define the control objective J_k^{acc} as:

$$J_k^{\text{acc}} = \varsigma_k \max\{0, B_k^{\text{acc}}\} \quad (24)$$

which is only active if lane-changing is chosen, i.e., $\varsigma_k = 1$.

6) *MPC Formulation*: We can summarize the aforementioned discussion and formulate the embedded optimization problem in MPC at time step k as follows:

$$\min_{U_k, X_k, Z_k, \varsigma_k} \text{Eq.(5)}$$

$$\text{s.t. Eqs.(7)-(13), Eqs.(17)-Eqs.(18),}$$

$$x(k|k) = \hat{x}(k) \quad (25)$$

$$\varsigma_k \in \{0, 1\} \quad (26)$$

where Eq.(25) sets the initial condition as the measured state $\hat{x}(k)$ at time step k , and Eq.(26) defines the lane-changing decision for the planning horizon as a binary variable. Note that the above optimization problem is a quadratic optimization problem for each ς_k (0 or 1), which can be easily solved with the barrier method.

B. HV Trajectory Prediction

Note that the MPC-based controller presented in Section IV-A (i.e., Eq.(15)) requires the trajectories of surrounding vehicles as input, which arises from various HDV behaviors such as car-following and lane-changing [40], [41]. To provide such input, we present an HV

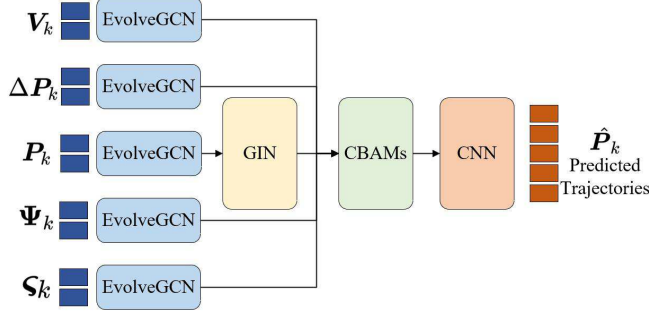


Fig. 3. Framework of the trajectory prediction algorithm.

trajectory prediction algorithm to predict the trajectories of the surrounding HVs over the planning horizon. Although it has been demonstrated that Graph Neural Networks (GNNs) can yield supreme performance in comparison with other algorithms [42], [43], [44], existing methods typically assume fixed graph topologies (i.e., the spatial relationship between vehicles). However, in weaving sections, intensive lane changes make the spatial relationship between vehicles highly dynamic, which can significantly impact the performance of existing methods. To address this issue, we incorporate the information contained in the evolution of graphs into existing GNN-based trajectory prediction algorithms via EvolveGCN, a neural network architecture to handle dynamic graphs. Specifically, we combine EvolveGCN with a universal trajectory prediction network called Pishgu [43], thanks to its satisfactory performance and simplicity of implementation.

At each time step k , SV (CAV with index i) collects information from surrounding HVs, denoted by $\mathcal{V} \subseteq \mathcal{M}$. Here, we notice that $\mathcal{M}$ contains both CAVs and HVs, while $\mathcal{V}$ consists of only HVs. Let a complete graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ denote the graph representing the topology of the surrounding HVs. Note that for brevity, we omit the subscript i of SV in this subsection, if such omission does not cause ambiguity.

In graph $\mathcal{G}$, the features of each HV $j \in \mathcal{V}$, denoted by $\eta_{j,k}$, contain five channels and span the previous T_{in} time steps, including HV j 's absolute position $\mathbf{P}_{j,k} = \{\mathbf{p}_j^{k-h}\}_{h=0}^{T_{in}-1}$, relative position to SV $\Delta\mathbf{P}_{j,k} = \{\Delta\mathbf{p}_j^{k-h}\}_{h=0}^{T_{in}-1}$, driving speed $\mathbf{V}_{j,k} = \{v_j^{k-h}\}_{h=0}^{T_{in}-1}$, heading angle $\Psi_{j,k} = \{\psi_j^{k-h}\}_{h=0}^{T_{in}-1}$, and lane change intention $\mathbf{s}_{j,k} = \{s_j^{k-h}\}_{h=0}^{T_{in}-1}$. Notice that the lane change intention of surrounding HVs can be inferred from the state of the turn signal. The feature set of all HV neighbors of SV, η_k , can be represented by $\eta_k = \{\eta_{j,k}\}_{j \in \mathcal{V}}$. The prediction algorithm predicts the future trajectory of all the nodes in graph $\mathcal{G}$, which is denoted as $\hat{\mathbf{P}}_k = \{\hat{\mathbf{P}}_{j,k}\}_{j \in \mathcal{V}}$, where $\hat{\mathbf{P}}_{j,k} = \{\hat{\mathbf{p}}_j^{k+h}\}_{h=1}^P$ within the prediction horizon of MPC.

Fig. 3 shows the framework of the trajectory prediction algorithm, which comprises four modules with different architectures: EvolveGCN, Graph Isomorphism Network (GIN), Convolutional Block Attention Modules (CBAM), and CNN. Note that in addition to the original Pishgu model solely relying on GIN, CBAM, and CNN, we integrate EvolveGCN to account for graph evolution.

(i) *EvolveGCN*. We design an EvolveGCN-based input layer to preprocess the features of all HV consisting of five channels,

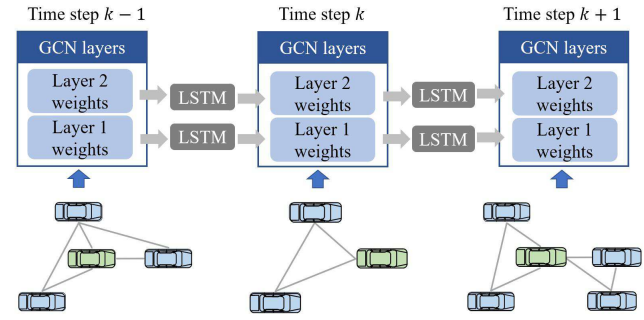


Fig. 4. Principle diagram of EvolveGCN layer. The green vehicle represents SV and blue vehicles indicates SV's surrounding HVs within the communication range.

i.e., $\mathbf{V}_k, \Delta\mathbf{P}_k, \mathbf{P}_k, \Psi_k, \mathbf{s}_k$, to capture the evolution of graph $\mathcal{G}$. As depicted in Fig. 4, EvolveGCN dynamically captures the evolution of graph topology by employing a Long Short-Term Memory (LSTM) to iteratively adjust the parameters of the Graph Convolutional Network (GCN). In EvolveGCN, the weight matrix of GCN recursively serves as both the input and the output of the LSTM over time. This allows for the direct adaptation of the GCN model across the temporal dimension without relying on the assumption of a fixed number of nodes, while also considering the evolution of the graph topology through the updating of the GCN weight matrix.

(ii) *GIN*. After feeding the features to EvolveGCN, we use GIN to further process the feature of absolute positions $\mathbf{P}_k$ to capture the strong temporal correlation between past absolute positions and future absolute positions. The benefit of using GIN instead of GCN is that by introducing self-loop edges into graph convolution, GIN can emphasize and prioritize the ego node's position over information from surrounding nodes. Note that GIN processes only the feature of absolute positions to highlight the temporal correlation between past and future absolute positions. This design allows the absolute position to play a dominant role in the feature processing process.

(iii) *CBAM*. The CBAM layer comprises a channel-spatial attention layer followed by a succeeding CNN layer. Here, the attention mechanism enables the CNN layer to dynamically determine the importance and contribution of the preceding modules' output features to the final prediction. Specifically, channel attention identifies important channels (e.g., absolute positions, speed, etc.), and spatial attention locates the important nodes of the graph. In the established model, the CBAM module is designed to consist of three sequential CBAM layers.

(iv) *CNN*. Finally, the output of the last CBAM layer passes through a conventional CNN layer to generate the final predicted trajectories $\hat{\mathbf{P}}_k$. Here, CNN is chosen for its compatibility and high performance in various prediction tasks with multimodal features [45], [46].

We train the neural network by minimizing the l_2 -loss, written as

$$\mathcal{L}_2(\tilde{\mathbf{P}}_k, \hat{\mathbf{P}}_k) = \|\tilde{\mathbf{P}}_k - \hat{\mathbf{P}}_k\|_2^2 \quad (27)$$

where $\tilde{\mathbf{P}}_k$ represents the real trajectory of SV's surrounding HVs in the training data.

V. SIMULATION

In this section, we perform simulations to evaluate the proposed bi-level controller in a typical weaving section. Section V-A introduces the benchmark algorithms, Section V-B describes the simulated scenario, Section V-C evaluates the performance of the proposed algorithm, Section V-D performs sensitivity analysis on weaving characteristics, including varied traffic inflow and segment length, and Section V-E performs sensitivity analysis to the cell size and upper-level decision-making frequency.

A. Benchmarks

The following control methods are compared to evaluate the proposed method:

- **HDM:** This benchmark, i.e., human driver model (HDM), implements a rule-based controller without coordination, which uses IDM for longitudinal control and MOBIL for lateral control. This benchmark resembles the driving behavior of HVs.
- **HLC-L:** This benchmark represents a rule-based controller with local coordination. Specifically, this benchmark replicates the work of [17], presenting a harmonious lane-changing (HLC) controller designed to coordinate vehicles within a limited range.
- **HLC-G:** This benchmark represents a rule-based controller with global coordination. Specifically, this benchmark is an extension of the work [17] with local coordination extended to the global scale, which is achieved by employing an EvolveGCN-based RL algorithm to centrally coordinate all vehicles in the weaving section.
- **MPC-Only:** This benchmark presents an MPC-based controller with no coordination, representing a trajectory planning-based approach implemented solely through our proposed lower-level MPC controller to enhance ego travel efficiency.
- **Bi-Level-L:** This benchmark presents an MPC-based controller with local coordination, which implements our proposed bi-level controller with only local information as in HLC-L controller.
- **Bi-Level-G:** This benchmark implements the complete version of our proposed bi-level controller, i.e., an MPC-based controller with global coordination.

The comparison between MPC, Bi-Level-L, and Bi-Level-G (as well as HDM, HLC-L, and HLC-G) illustrates the value of performing global coordination, while the comparison between MPC-based methods and HLC-G demonstrates the value of using lower-level MPC.

B. Implementation Details

We conduct a simulation-based case study [47], inspired by a weaving section near Basel, Switzerland, where the parameters are calibrated with empirical data on Nov. 3, 2014. The weaving section is 535 meters long and comprises two main lanes and one auxiliary lane. The simulation replicates a typical evening peak from 16:00 to 17:00, during which high

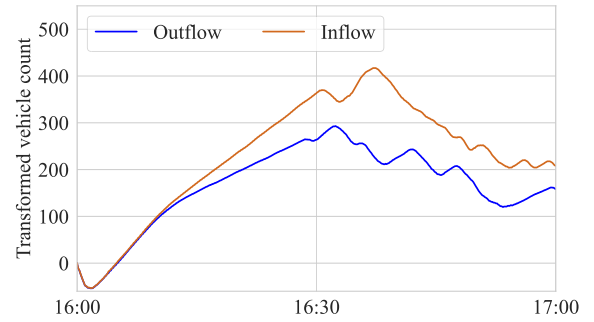


Fig. 5. Inflow and outflow of the studied area, where an initial capacity of 5400 veh/h is observed during 16:00–16:10, and a sustained capacity of 4080 veh/h is observed during 16:35–16:50, yielding a capacity drop of 24.4%.

traffic demand occurs and capacity drops are often observed. Fig. 5 illustrates the inflow and outflow of the studied area with a background flow of 4800 veh/h, which in reality features a clear capacity drop after the onset of congestion as the discharging flow reduces at 16:10. We set a time step of 0.2 s.

In the upper-level RL controller, we divide each lane of the weaving section evenly into 20 cells. Hence, the cell representation $\mathcal{N}$ of the upper-level RL controller is configured as a $3 \times 60 \times 5$ tensor. These three dimensions correspond to the feature dimension, the number of cells, and the length of historical information, respectively. In our reward function, $q_{\max}$ is set to 1800 veh/(lane · h) following the setting in [48]. We also set $\beta^{\text{spd}} = 1$, $\beta^{\text{flow}} = 1$, $\beta^{\text{miss}} = -2$ and $\beta^{\text{safety}} = -1$ following the recommendations in [17] and further refined them based on our simulated scenario. Moreover, the actor network of PPO includes two CNN layers, succeeded by three fully connected layers with 128, 512, and 384 neurons, respectively. An output layer with 300 neurons is added to produce actions for 60 cells. Similarly, the critic network of PPO includes two CNN layers, succeeded by three fully connected layers with 64, 128, and 64 neurons, respectively. The output dimension is set to 1 to produce an estimated scalar state value. The learning rates of actor and critic nets are both set to be 0.0003, which is an empirical value widely used in existing literature [49], [50].

In the lower-level MPC controller, the default values of control weights, α^{effort} , $\alpha^{\text{exit},x}$, $\alpha^{\text{exit},y}$, α^{vel} , and α^{acc} , are set to 0.05, 0.25, 0.5, 0.5, and 0.05, respectively, following the recommendations of empirical studies [51], [52] and are manually fine-tuned to suit the simulated scenario. Note that these default values are used only in the benchmark of MPC-Only, as the other benchmarks either not use. Notice that for bi-level controllers (i.e., Bi-Level-G and Bi-Level-L), these coefficients will be controlled by the upper-level controller. Moreover, the prediction horizon of the MPC controller is $P = 16$.

In the HV trajectory predictor, the length of the past observation is $T_{\text{in}} = 10$, following the original setting of Pishgu. The learning rate of the trajectory predictor is 0.0005. We generate the HV trajectory training dataset by randomly generating 100 simulation scenarios with different penetration rates of CAV of 100%, 80%, 60%, 40%, and 20%. Within each

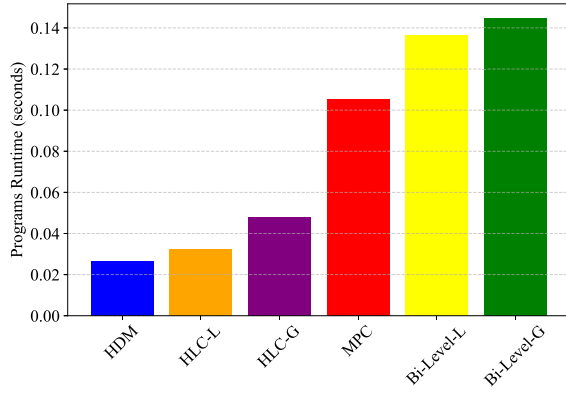


Fig. 6. Single-step computing time of benchmarks.

scenario, we collect the real-time locations, speeds, heading angles, and lane change control variables of each vehicle. In total, we collect 530,000 samples, which are then transformed into the training data of the neural networks.

Each trainable benchmark (Bi-Level-G, Bi-Level-L, HLC-G, and HLC-L) is trained for 50 episodes, during which they converge to optimal performance. Then an additional five episodes are conducted to test the control performance. The experiments were conducted partly on a computer equipped with an RTX 3070 GPU and partly on a high-performance computing platform featuring NVIDIA H100 and A40 GPUs. The designed neural networks and MPC control scheme are implemented in Python with Pytorch 1.13.1 and Pyomo 6.6.1. Fig. 6 compares the single-step computing time of all benchmarks. We can see that although MPC controllers impose a higher computational burden than rule-based controllers, they still meet real-time control requirements, with all computation times remaining below the 0.2-second simulation step. This efficiency is achieved by deliberately formulating the MPC problem as a quadratic programming problem, avoiding the substantial computational cost associated with solving nonconvex optimization problems. Additionally, we find that incorporating coordination through the upper-level RL controller does not significantly increase computational time, even in global coordination scenarios. This can be attributed to the high online computational efficiency of RL, as the complex optimization is primarily handled during the offline training phase.

C. Performance in Testing Scenarios

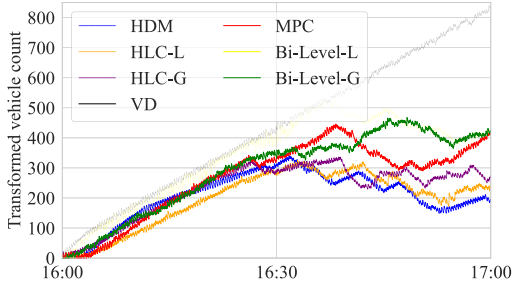
Table I and Fig. 7 show the performance of the benchmarks in scenarios with various CAV penetration rates, where the incoming flow to the weaving section is obtained from empirical data. Specifically, Table I summarizes the space-mean speed, average delay time and throughput (i.e., the total number of vehicles that leave the weaving section.) of the weaving section resulting from all benchmarks, and Fig. 7 demonstrates the queuing diagrams and fundamental diagrams resulting from all benchmarks. Fig. 8 and Fig. 9 compare the comfort level (measured by the mean absolute jerk in m/s^3) and the safety level (measured by the average Time-To-Collision (TTC) in seconds). Fig. 10 evaluates the resulting

TABLE I
PERFORMANCE OF BENCHMARK ALGORITHMS. THE VALUES IN PARENTHESES INDICATE THE PERCENTAGE OF IMPROVEMENT COMPARED TO THE HDM BENCHMARK. THE OPTIMAL ALGORITHM FOR EACH SCENARIO IS HIGHLIGHTED IN BOLD

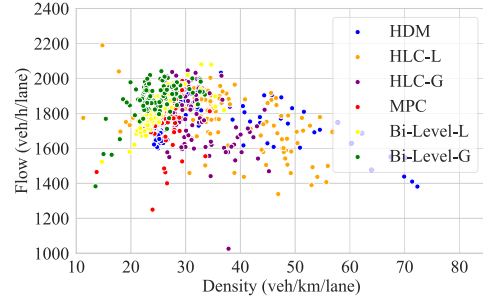
Penetration Rate	Benchmark	Space Mean Speed (km/h)	Average Delay Time (s)	Throughput (veh/(lane h))
0%	HDM	55.5	18.7	1663
20%	HLC-L	56.9 (2.6%)	17.8 (4.8%)	1677 (0.8%)
	HLC-G	57.0 (2.9%)	17.7 (5.2%)	1690 (1.6%)
	MPC	58.0 (4.5%)	17.2 (8.1%)	1737 (4.4%)
	Bi-level-L	58.1 (4.6%)	16.7 (10.6%)	1739 (4.6%)
	Bi-Level-G	59.4 (7.0%)	15.9 (15.1%)	1740 (4.6%)
40%	HLC-L	58.8 (6.0%)	16.7 (10.1%)	1682 (1.1%)
	HLC-G	59.0 (6.4%)	16.6 (11.2%)	1694 (1.9%)
	MPC	60.2 (8.6%)	15.9 (14.6%)	1762 (6.0%)
	Bi-level-L	61.5 (10.8%)	15.0 (19.6%)	1765 (6.1%)
	Bi-Level-G	62.5 (12.6%)	14.2 (24.0%)	1772 (6.6%)
60%	HLC-L	59.8 (7.8%)	16.2 (13.4%)	1687 (1.4%)
	HLC-G	60.3 (8.7%)	15.9 (14.9%)	1707 (2.6%)
	MPC	63.9 (15.2%)	14.1 (24.5%)	1774 (6.7%)
	Bi-level-L	64.7 (16.5%)	13.7 (26.6%)	1778 (6.9%)
	Bi-Level-G	65.9 (18.7%)	13.0 (30.6%)	1788 (7.5%)
80%	HLC-L	63.2 (14.0%)	14.4 (22.8%)	1735 (4.3%)
	HLC-G	64.9 (17.0%)	13.6 (26.9%)	1753 (5.4%)
	MPC	70.8 (27.7%)	11.2 (40.3%)	1804 (8.5%)
	Bi-level-L	71.0 (27.9%)	10.9 (41.4%)	1812 (8.9%)
	Bi-Level-G	73.8 (32.9%)	9.9 (47.0%)	1848 (11.1%)
100%	HLC-L	74.1 (33.6%)	9.9 (46.7%)	1790 (7.6%)
	HLC-G	76.2 (37.4%)	9.2 (50.5%)	1809 (8.8%)
	MPC	80.1 (44.5%)	8.0 (57.2%)	1860 (11.8%)
	Bi-level-L	82.8 (49.1%)	7.1 (61.9%)	1862 (12.0%)
	Bi-Level-G	83.6 (50.6%)	6.4 (65.7%)	1873 (12.6%)

capacities from all benchmarks in scenarios with various CAV penetration rates, which are obtained by increasing the incoming flow to the weaving section until a sustainable outflow can be measured.

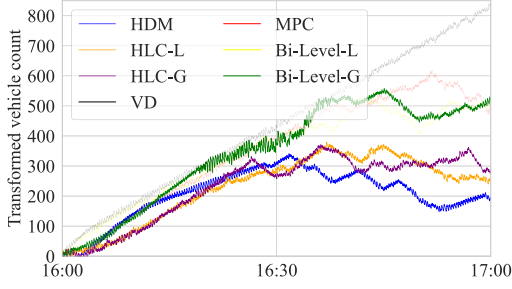
Value of global coordination: We show the value of coordination by comparing Bi-Level-G with Bi-Level-L and MPC controllers. First, from Table I, we can see a consistent improvement in space-mean speed and throughput, along with a continuous reduction in average delay time, as the level of coordination increases. Specifically, Bi-Level-G consistently achieves the highest space-mean speed and throughput while maintaining the lowest delay time across various CAV penetration rates, outperforming both Bi-Level-L and MPC controllers. Furthermore, Bi-Level-L also demonstrates superior performance compared to the MPC controller. Second, we can see from the queuing diagrams (i.e., the left column) of Fig. 7 that Bi-Level-G reduces the magnitude of the capacity drop in all scenarios tested, especially in scenarios with low penetration rates (e.g., 20% as in Fig. 7a). Notice that under 100% CAV penetration rate, the cumulative curve for Bi-Level-G almost completely overlaps the virtual departure curve, demonstrating that our coordination algorithm significantly mitigates congestion. We can also see from the fundamental diagrams (i.e., the right column) of Fig. 7 that Bi-Level-G improves the slope of the free-flow branch, i.e., enabling a higher driving speed. Third, we can see from Fig. 10 that the capacity increases as the level of coordination increases from no coordination to global



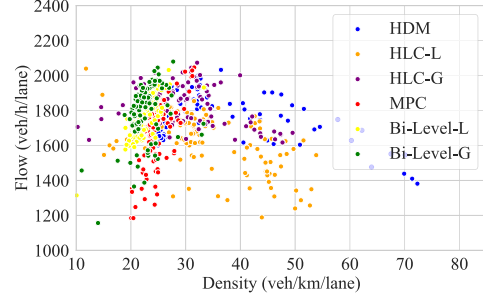
(a) QD under 20% CAV penetration rate.



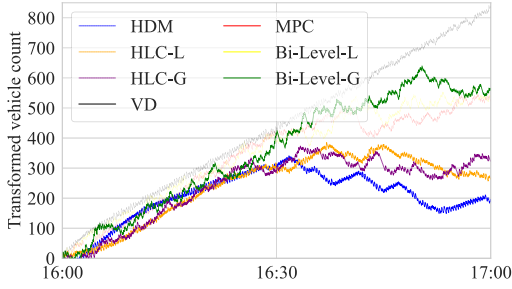
(b) FD under 20% CAV penetration rate.



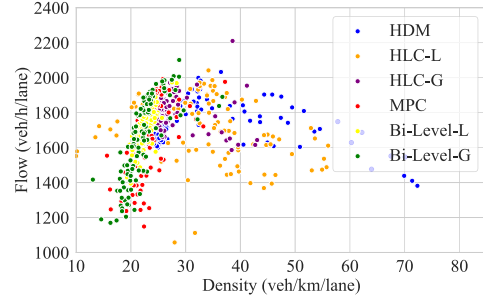
(c) QD under 40% CAV penetration rate.



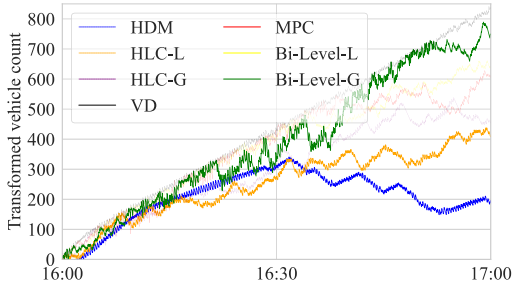
(d) FD under 40% CAV penetration rate.



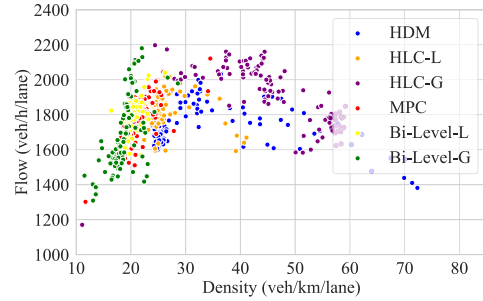
(e) QD under 60% CAV penetration rate.



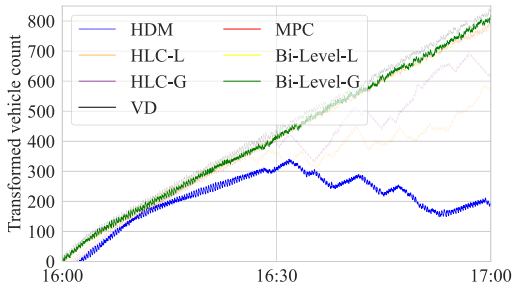
(f) FD under 60% CAV penetration rate.



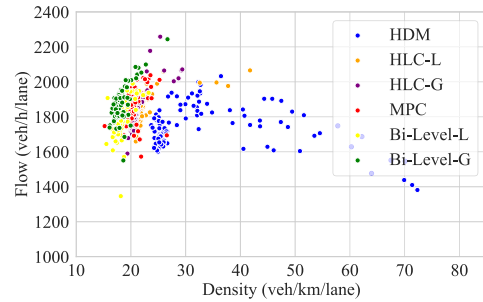
(g) QD under 80% CAV penetration rate.



(h) FD under 80% CAV penetration rate.



(i) QD under 100% CAV penetration rate.



(j) FD under 100% CAV penetration rate.

Fig. 7. Queuing diagrams (QDs) with background flow of 4800 veh/h and fundamental diagrams (FDs) under different CAV penetration rates. Notice that VD represents the virtual departure curve at the exit of the weaving section, which is obtained from the simulated traffic demand based on the empirical data.

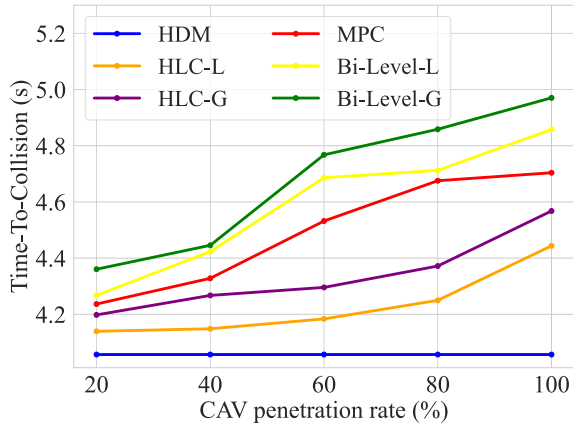


Fig. 8. Time-To-Collision comparison under different CAV penetration rates.

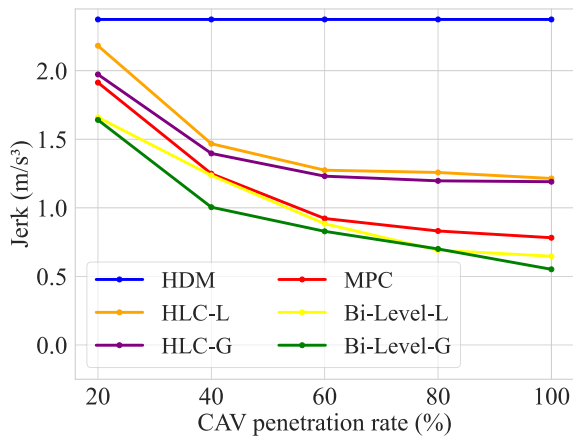


Fig. 9. Jerk comparison under different CAV penetration rates.

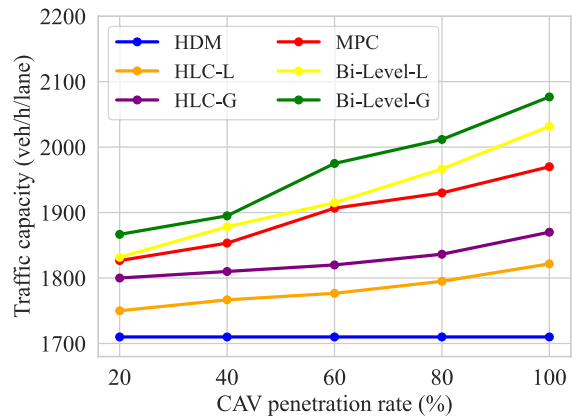


Fig. 10. Traffic capacity comparison under different CAV penetration rates.

coordination, i.e., Bi-Level-G yields a higher capacity than both Bi-Level-L and MPC. Similar conclusions can also be drawn from rule-based controllers by comparing HDM, HLC-L, and HLC-G.

Value of Lower-level Controller: We assess the value of our lower-level MPC controller by comparing the MPC controller with the HLC-G controller. As shown in Table I, it is evident that all MPC-based controllers (Bi-Level-G, Bi-Level-L and MPC) outperform the rule-based controllers (HLC-G,

TABLE II
TRAJECTORY PREDICTION ACCURACY (METER, IN UNITS OF 10^{-1})

Time Step	X-axis			Y-axis		
	Pishgu	GCN-Pishgu	Ours	Pishgu	GCN-Pishgu	Ours
1	2.53	2.06	1.91	0.16	0.14	0.12
2	2.60	2.18	2.27	0.16	0.14	0.12
3	2.88	2.19	2.14	0.16	0.14	0.12
4	2.88	2.31	2.01	0.16	0.14	0.12
5	3.01	2.45	2.16	0.16	0.14	0.12
6	3.16	2.77	2.50	0.16	0.14	0.12
7	3.87	3.45	3.11	0.16	0.14	0.13
8	4.77	4.34	3.89	0.17	0.14	0.13
9	5.86	5.43	4.93	0.17	0.15	0.13
10	7.14	6.69	6.11	0.17	0.15	0.13
11	8.57	8.11	7.52	0.17	0.15	0.13
12	10.17	9.68	9.07	0.17	0.15	0.13
13	11.91	11.39	10.82	0.17	0.15	0.14
14	13.73	13.22	12.56	0.17	0.16	0.14
15	15.70	15.16	14.53	0.18	0.16	0.14
16	17.72	17.18	16.61	0.19	0.17	0.15

HLC-L and HDM) in terms of space-mean speed, delay time and throughput. Fig. 8 and Fig. 9 show that the MPC-based controller achieves a higher Time-To-Collision and improved comfort level (i.e., lower jerk), which can be attributed to the advantage of multi-objective optimization in MPC, allowing it to balance multiple goals such as safety, efficiency, and comfort. We can also see from Fig. 7 and Fig. 10 that MPC-based approaches significantly reduce the magnitude of the capacity drop, achieve a higher driving speed, and improve capacity, compared to rule-based controllers. Further note that the transformed cumulative curves for some benchmarks may temporarily fall below that of HDM. This occurs because different algorithms create bottlenecks at different times, leading to occasional performance fluctuations. However, these variations do not affect overall performance, as all benchmarks ultimately surpass pure HV by the end of the simulation in terms of the throughput.

Value of CAV Penetration Rate: We evaluate the value of the CAV penetration rates by comparing benchmark performance under different CAV penetration rates. As demonstrated in Table I, the improvement of Bi-Level-G in terms of space-mean speed, delay time and throughput increases as the CAV penetration rate increases. Specifically, the throughput can be increased by up to 12.6%, the delay time can be reduced by up to 65.7%, and the space-mean speed can be increased by up to 50.6%. Fig. 8 and Fig. 9 show that as the CAV penetration rate increases, both safety and comfort level of vehicles are improved. Moreover, from Fig. 10, we can see a notable increase in capacity as CAV penetration improves, which is valid for any benchmark.

Value of EvolveGCN in Trajectory Prediction: We evaluate the benefits of incorporating the EvolveGCN layer in the trajectory predictor by comparing the resulting Root Mean Square Errors (RMSEs) of (1) the Pishgu algorithm, (2) the Pishgu algorithm with GCN to handle the spatial correlation between vehicles, and (3) our proposed trajectory prediction algorithm. As shown in Table II, our predictor outperforms the original Pishgu and GCN-Pishgu by explicitly modeling

TABLE III

PERFORMANCE OF BENCHMARK ALGORITHMS. THE RATIO OF THROUGH, DIVERGING, AND MERGING FLOWS IS 1.00 : 0.69 : 0.53

Penetration Rate	Benchmark	Space Mean Speed (km/h)	Average Delay Time (s)	Throughput (veh/(lane h))
0%	HDM	51.2	21.6	1596
20%	HLC-L	52.8 (3.1%)	20.4 (5.6%)	1651 (3.4%)
	HLC-G	53.3 (4.1%)	20.1 (6.9%)	1662 (4.1%)
	MPC	54.9 (7.2%)	19.1 (11.6%)	1685 (5.6%)
	Bi-level-L	56.2 (9.8%)	18.2 (15.7%)	1701 (6.6%)
	Bi-Level-G	57.3 (11.9%)	17.5 (18.9%)	1717 (7.6%)
60%	HLC-L	56.2 (9.8%)	18.2 (15.7%)	1684 (5.5%)
	HLC-G	57.4 (12.1%)	17.5 (18.9%)	1696 (6.3%)
	MPC	59.9 (17.0%)	16.1 (25.5%)	1746 (9.4%)
	Bi-level-L	62.7 (22.5%)	14.7 (31.9%)	1772 (11.0%)
	Bi-Level-G	63.2 (23.4%)	14.4 (33.3%)	1782 (11.7%)

dynamic graph topology. Compared to the original Pishgu, employing Graph Convolutional Networks improves prediction performance by leveraging vehicle distribution topology to capture spatial relationships and enhance environmental perception. However, frequent lane-changing behaviors make vehicle interactions highly dynamic. To address this, we use EvolveGCN, which adapts to real-time topological changes by evolving GCN parameters through an LSTM layer, further reducing trajectory prediction errors.

D. Sensitivity Analysis to Weaving Characteristics

In this subsection, we conduct a sensitivity analysis to evaluate the impact of different weaving section characteristics, including varied traffic inflow ratios and segment lengths. Specifically, we compare space-mean speed, average delay time, and throughput under two representative CAV penetration rates (20% and 60%).

Sensitivity analysis on inflow ratios: We vary the inflow ratios quantified by ratios between different origin-destination (OD) pairs, i.e., mainline to mainline (through), mainline to off-ramp (diverging), and on-ramp to mainline (merging). Specifically, we evaluate two OD scenarios in terms of through, diverging, and merging flows: 1.00 : 0.69 : 0.53 and 1.00 : 0.45 : 0.36 (the original ratio is 1.00 : 0.56 : 0.44).

Tables III and IV show that the proposed Bi-Level-G method remains robust and effective under both OD ratio scenarios. It consistently achieves substantial improvements in space mean speed, delay reduction, and throughput across all CAV penetration rates. These performance gains are particularly pronounced under high-demand conditions, where lane-changing interactions are more frequent and complex.

Sensitivity analysis on weaving segment lengths: We further evaluate the performance of our method in scenarios with longer (635m) and shorter (435m) weaving segments, compared to the conventional length of 535m.

Tables V and VI show that the proposed method maintains superior performance in both shorter and longer weaving segments, demonstrating its robustness across varying road lengths. Despite the challenges posed by limited

TABLE IV

PERFORMANCE OF BENCHMARK ALGORITHMS. THE RATIO OF THROUGH, DIVERGING, AND MERGING FLOWS IS 1.00 : 0.45 : 0.36

Penetration Rate	Benchmark	Space Mean Speed (km/h)	Average Delay Time (s)	Throughput (veh/(lane h))
0%	HDM	61.4	15.3	1726
20%	HLC-L	64.1 (4.4%)	14.0 (8.5%)	1744 (1.0%)
	HLC-G	65.3 (6.4%)	13.4 (12.4%)	1761 (2.0%)
	MPC	66.2 (7.8%)	13.1 (14.4%)	1785 (3.4%)
	Bi-level-L	66.9 (9.0%)	12.7 (17.0%)	1797 (4.1%)
	Bi-Level-G	68.3 (11.2%)	12.1 (20.9%)	1813 (5.0%)
60%	HLC-L	67.7 (10.3%)	12.4 (19.0%)	1774 (2.8%)
	HLC-G	69.2 (12.7%)	11.8 (22.9%)	1782 (3.2%)
	MPC	70.9 (15.5%)	11.1 (27.5%)	1810 (4.9%)
	Bi-level-L	72.2 (17.6%)	10.6 (30.7%)	1837 (6.4%)
	Bi-Level-G	75.6 (23.1%)	9.4 (38.6%)	1869 (8.3%)

TABLE V

PERFORMANCE OF BENCHMARK ALGORITHMS. THE WEAVING SEGMENT IS INCREASED TO 635 METERS

Penetration Rate	Benchmark	Space Mean Speed (km/h)	Average Delay Time (s)	Throughput (veh/(lane h))
0%	HDM	59.4	19.4	1707
20%	HLC-L	60.5 (1.9%)	18.7 (3.6%)	1738 (1.8%)
	HLC-G	61.8 (4.0%)	17.9 (7.7%)	1742 (2.1%)
	MPC	63.3 (6.6%)	17.0 (12.4%)	1766 (3.5%)
	Bi-level-L	64.9 (9.3%)	16.1 (17.0%)	1789 (4.8%)
	Bi-Level-G	65.7 (10.6%)	15.7 (19.1%)	1803 (5.6%)
60%	HLC-L	64.1 (7.9%)	16.6 (14.4%)	1742 (2.1%)
	HLC-G	65.8 (10.8%)	15.6 (19.6%)	1774 (3.9%)
	MPC	68.3 (15.0%)	14.4 (25.8%)	1795 (5.2%)
	Bi-level-L	70.7 (19.0%)	13.2 (32.0%)	1814 (6.3%)
	Bi-Level-G	71.8 (20.9%)	12.7 (34.5%)	1824 (6.9%)

TABLE VI

PERFORMANCE OF BENCHMARK ALGORITHMS. THE WEAVING SEGMENT IS DECREASED TO 435 METERS

Penetration Rate	Benchmark	Space Mean Speed (km/h)	Average Delay Time (s)	Throughput (veh/(lane h))
0%	HDM	52.9	16.5	1610
20%	HLC-L	54.2 (2.5%)	15.9 (3.6%)	1624 (0.9%)
	HLC-G	56.3 (6.4%)	14.7 (10.9%)	1637 (1.7%)
	MPC	57.4 (8.5%)	14.2 (13.9%)	1678 (4.2%)
	Bi-level-L	57.8 (9.3%)	14.0 (15.2%)	1722 (7.0%)
	Bi-Level-G	59.2 (11.9%)	13.4 (18.8%)	1724 (7.1%)
60%	HLC-L	57.3 (8.3%)	14.2 (13.9%)	1632 (1.4%)
	HLC-G	59.7 (12.9%)	13.1 (20.6%)	1655 (2.8%)
	MPC	61.5 (16.3%)	12.4 (24.8%)	1711 (6.3%)
	Bi-level-L	63.7 (20.4%)	11.5 (30.3%)	1748 (8.6%)
	Bi-Level-G	64.3 (21.6%)	11.3 (31.5%)	1767 (9.8%)

maneuvering space in short segments and the increased coordination demands in longer segments, our method consistently outperforms other benchmarks.

E. Sensitivity Analysis to Algorithmic Parameters

In this subsection, we perform sensitivity analysis on the cell size and the upper-level decision-making frequency.

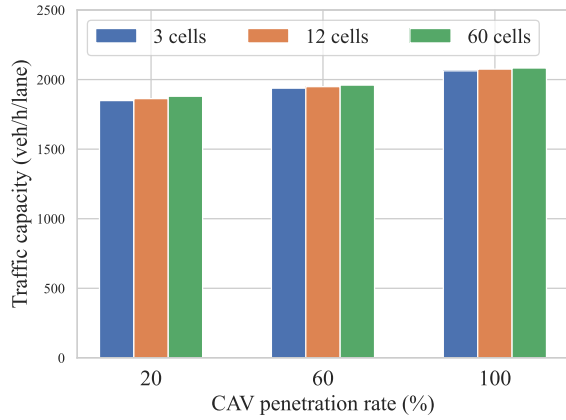


Fig. 11. Traffic capacity comparison of Bi-Level-G under different cell sizes.

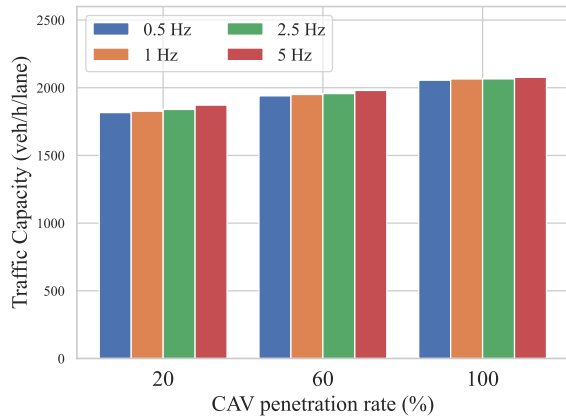


Fig. 12. Traffic capacity comparison of Bi-Level-G under different upper-level decision-making frequencies.

We specifically compare the resulting traffic capacity from various levels of cell sizes and upper-level decision-making frequencies in scenarios with three typical CAV penetration rates (20%, 60%, and 100%).

Sensitivity analysis on cell size: As depicted in Fig. 11, we observe that increasing the cell size slightly improves traffic capacity for all CAV penetration rates. This is because a more refined cell representation enables greater flexibility in tuning control weights. Nevertheless, the improvement is not significant, indicating that we can achieve the benefits of global coordination with a coarse cell representation.

Sensitivity analysis on upper-level decision-making frequency: Fig. 12 illustrates the traffic capacity under various decision-making frequencies at the upper level: 0.5Hz, 1Hz, 2.5Hz, and 5Hz. The result shows that as the decision-making frequencies at the upper level increases, the traffic capacity improves accordingly in all CAV penetration rates. By increasing the upper-level decision-making frequencies from 0.5Hz to 5Hz, the capacity can increase by up to 3.02%, which indicates that our proposed framework can be used even if the traffic operators and CAVs are making decisions at different frequencies.

VI. CONCLUSION

In this paper, we propose a bi-level controller to implement global coordinated lane change control in mixed-traffic weaving sections. The upper-level RL-based controller perceives vehicular information from all vehicles and then determines control weights to balance multiple objectives in the lower-level MPC-based controller. The lower level takes the upper level's decision and solves an embedded optimization problem for each individual vehicle to obtain a decision corresponding to the vehicle's local situation. It is worth noting that the proposed bi-level framework is general and can be easily applied and adapted to weaving sections of various forms, regardless of their length or structural complexity. The generalizability results from the rationale behind the bi-level design, whereby the invariant upper-level controller (vehicle-side) ensures broad applicability, and the upper-level controller (road-side) can be fine-tuned online to address scenario-specific variations. Moreover, to enable the deployment of our bi-level controller in mixed traffic with unknown HV behavior, we propose an HV trajectory prediction algorithm at the lower level, where EvolveGCN is incorporated to consider the evolution of multiple HVs around a CAV. The results demonstrate that the proposed algorithm surpasses state-of-the-art baselines, increasing driving speed and capacity by up to 50.6% and 12.6%, respectively.

This research opens several interesting directions. First, we aim to improve the generalizability of the proposed approach to various weaving sections, other highway scenarios (e.g., merges and diverges), and weather conditions, where transfer learning or meta-learning can be applied to accelerate the re-training process during deployment. Second, we would like to combine our proposed approach with other control methods, such as ramp metering and variable speed limits for inbound vehicles.

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